ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** A is true but R is false.	A is true; R is false because animal cells lack a cell wall.
2	**(b)** 2 mm	2 x 1000 = 2000 micrometres = 2 mm.
3	**(b)** stops functioning properly	Nucleus controls the cell.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Chloroplasts carry out photosynthesis, which is why plants (which make their own food) have them.
5	**(c)** 0.05 mm	5 / 100 = 0.05 mm.
6	**(c)** yeast is more active at warm temperature	Warmth boosts fermentation.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** makes food by photosynthesis	Chloroplasts trap light for photosynthesis.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The compartments he saw were the walls of dead plant cells, which is why they looked like small rooms.
10	**(a)** Both A and R are true and R is the correct explanation of A.	The cell wall gives the plant cell its fixed shape.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Actual = 5 / 500 = 0.01 mm (1); = 10 micrometres (1).	1+1
12	Amoeba has only a flexible membrane (1); Plant cell has a rigid wall (1).	1+1
13	All organisms are built of cells (1); Cells combine into tissues, organs and bodies (1).	1+1
14	Cork consists of dead cells (1); Only the walls remain after the contents are lost (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Onion cells are rectangular with a cell wall; cheek cells are rounded or irregular without a wall (1); Both show a nucleus and cytoplasm; the onion cell has a large vacuole (1); Stains such as safranin or methylene blue make the parts visible (1).	1+1+1
16	Digestion: food vacuole (1); Excretion: contractile vacuole (1); Movement: pseudopodia (1).	1+1+1
17	Antibiotics act on bacteria (1); Fungi have different cell walls (1); Fungicides are needed (1).	1+1+1
	OR: Virus is smaller than bacterium (1); Virus is non-cellular; bacterium is a cell (1); Bacterial diseases need antibiotics; viral diseases need vaccines/antivirals (1).	1+1+1
18	No division of labour (1); Complex functions cannot be done (1); Organisms would stay simple (1).	1+1+1
	OR: Egg cell is about 0.1 mm (1); Bacterium is about a micrometre (1); A cell's size depends on its function (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A thin peel lets light pass through so one layer of cells can be seen (1). (ii) Nucleus (1). (iii) Cell wall (1). (iv) Cheek cells have no cell wall and are irregular/rounded in shape, and have no large central vacuole (any two) (2).	1+1+1+2
20	Objective = 800 / 10 = 80x (1); Actual = 4 / 800 mm (1); = 0.005 mm (1); = 5 micrometres (1); This size suggests a bacterium (1).	1+1+1+1+1
	OR: Milk is warmed to lukewarm (1); A spoon of curd containing Lactobacillus is added (1); Bacteria multiply and change lactose to lactic acid (1); Milk sets as curd (1); Useful for digestion (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.

- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.
- Forgetting the stain, thin peel and cover slip when describing slide preparation.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** virus, bacterium, red blood cell	Smaller to larger.
2	**(d)** Flu is caused by a virus	Viruses are not affected.
3	**(d)** A is false but R is true.	A is false: the cells were dead. R is true.
4	**(c)** specialised for movement	Tail helps in swimming.
5	**(c)** it has a rigid cell wall	Wall fixes the shape.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Reason explains it.
7	**(b)** cell size varies widely among organisms	Sizes range from micrometres to centimetres.
8	**(d)** wall is outside the membrane	Wall is outermost.
9	**(d)** A is false but R is true.	Animal cells do not have chloroplasts, so A is false; R is true.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Most cell parts are nearly colourless; the stain provides contrast.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pathogen: disease-causing microbe, e.g. typhoid bacterium (1); Decomposer: recycles dead matter, e.g. soil fungi (1).	1+1
12	Many are harmless or helpful (1); Some make vitamin K and help in digestion (1).	1+1
13	Antibiotics act on bacterial cell structures (1); Viruses have no cells and multiply only inside host cells (1).	1+1
14	Yeast ferments sugar (1); Releases carbon dioxide that fills the balloon (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nucleus: manager (1); Cytoplasm: working floor (1); Membrane: gate (1).	1+1+1
16	They have different functions (1); Different parts become active (1); Cells specialise during development (1).	1+1+1
17	Stain not used or not enough (1); Cells are folded (1); Wrong focus (1).	1+1+1
	OR: Dead cells lose their living contents (1); Walls remain to give support (1); Such cells occur in xylem and cork (1).	1+1+1
18	Unicellular divide into two (1); Multicellular produce gametes/seeds (1); Multicellular are more complex (1).	1+1+1
	OR: Large surface area compared to volume (1); Materials move in and out quickly (1); Cell activities are efficient (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The cell (1). (ii) Unicellular (1). (iii) Multicellular (1). (iv) Different cells are specialised for different jobs (e.g. nerve cells, muscle cells), so division of work makes the organism function efficiently (2).	1+1+1+2
20	Nerve: long, branched (1); RBC: disc, no nucleus (1); Root hair: long projection (1); Muscle: spindle, contract (1); Guard cell: bean-shaped, control pores (1).	1+1+1+1+1
	OR: Wall: support (1); Membrane: control (1); Vacuole: storage (1); Wall is porous (1); Membrane is selective (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.
- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.
- Forgetting the stain, thin peel and cover slip when describing slide preparation.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** bigger organisms have more cells, not bigger cells	Body size depends on the number of cells.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Shape suits function.
3	**(a)** virus	Viruses are too small.
4	**(d)** lacks chlorophyll and cannot make food	Fungi are heterotrophic.
5	**(c)** A is true but R is false.	A is true; R is false.
6	**(c)** organ it belongs to	Organs depend on tissues.
7	**(c)** cause disease in host organisms	Disease-causing.
8	**(c)** A is true but R is false.	A is true; R is false as fungi lack chlorophyll.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but being single-celled does not make them prokaryotic; the lack of a membrane-bound nucleus does.
10	**(c)** A is true but R is false.	A is true; R is false because bacterial cells have no true nucleus.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Organelle: part inside a cell (1); Organ: part of a body made of tissues (1).	1+1
12	Wall is permeable; does not control materials (1); Membrane regulates entry and exit (1).	1+1
13	Stain brings out the cell parts (1); Cover slip keeps the specimen flat and protected (1).	1+1
14	Leaves get light (1); They are the site of photosynthesis (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	20 micrometres = 0.02 mm (1); 0.07 / 0.02 (1); = 3.5, about 3 to 4 cells (1).	1+1+1
16	No wall: draw outermost wall (1); Add a large central vacuole (1); Also label chloroplasts and nucleus (1).	1+1+1
17	Do not touch the lens (1); Handle slides gently (1); Do not use direct sunlight (1).	1+1+1
	OR: Number of doublings = 120 / 20 = 6 (1); 2^6 (1); = 64 bacteria (1).	1+1+1
18	Place a drop of curd water or pond water (1); Stain with methylene blue (1); Use the highest power (1).	1+1+1
	OR: Showed that germs cause disease (1); Led to cleanliness, boiling and sterilisation (1); Vaccines were developed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 10 x 40 = 400 times (1). (ii) The red blood cell (1). (iii) Vacuole (1). (iv) A virus is far smaller than a bacterium and is below the limit of a light microscope; a much more powerful electron microscope is needed (2).	1+1+1+2
20	Bacteria: no nucleus; typhoid (1); Fungi: wall, no chlorophyll; ringworm (1); Protozoa: with nucleus; malaria (1); Size ranges differ (1); All can be seen under a microscope (1).	1+1+1+1+1
	OR: Leeuwenhoek: single lens (1); Hooke: compound microscope (1); Greater magnification followed (1); Light microscope limit: cannot see viruses (1); Electron microscopes see viruses (1).	1+1+1+1+1

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- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
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